

Comparative study on electromagnetohydrodynamic single-wall carbon nanotube–water dusty nanofluid in the presence of radiation and Ohmic heating

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Abstract

The main intention of this article is to explore the significance of electromagnetohydrodynamic (EMHD) dusty nanofluid on the stagnation point flow in the presence of radiation and Ohmic heating. Water is considered as base fluid with single-wall carbon nanotubes along with suspended dust particles. The numerical solutions for both dusty phase and fluid phase differential systems are obtained by using Runge Kutta 4th order with shooting method. Influence of embedded parameters such as Biot number, electric field, viscous dissipation, magnetic field, thermal radiation, Joule heating and stagnation point flow are presented graphically. Dust particles play a significant role in food pharmaceutical industries particularly for efficient cultivation of microorganisms and also these type of studies may be applicable in atmospheric fields, wastewater treatment, metal objects covered with plastic materials and solids drying. The higher values of electric field parameter (E) increases the fluid phase velocity and dusty phase velocity. An increase in the thermal radiation parameter releases thermal energy into the flow, so this energy helps to increase the fluid phase temperature and dusty phase temperature. The fluid phase temperature and dusty phase temperature declines with increasing the electric field parameter for dusty fluid and dusty nanofluid cases.

Keywords

Electromagnetohydrodynamic, stagnation point, single-wall carbon nanotube, dusty nanofluid, viscous dissipation, Joule heating

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Introduction

A dusty fluid is an alloying of millimeter (or) micrometer-sized particles (dust particles) into the base fluid. Several researchers are investigating the flow of dusty fluid due to its recent developments and importance in industries, engineering and technology demands. In recent days dusty fluid has a wide range of applications like atmospheric fields, wastewater treatment, Metal objects covered with plastic materials and solids drying, recovery of crude oil from pores of reservoir rocks and purification, soil salvation by natural winds, power plant piping, conveying of powdered materials, Coal gasification for power generation, Production of granular polyethylene, food pharmaceutical industries particularly for efficient cultivation of microorganisms, lunar surface erosion by the exhaust of a landing vehicle,

fluidized beds and fluid droplet sprays.^{1,2} Souayah et al.³ presented a steady two-dimensional boundary layer flow and heat transfer rate over the stretching surface of an incompressible dusty hybrid nanofluid. Hady et al.⁴ studied the numerically 2D laminar natural convection flow of a micropolar dusty fluid with convective boundary conditions over a stretching surface. Gireesha et al.⁵ presented the boundary layer

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flow of a dusty Casson fluid over a stretching sheet along with Cattaneo Christov heat flux model.

In recent years the effect of a magnetic field and electric fields has remarkable attention. Many researchers studied the effect of a magnetic field, and they refuse the induced magnetic field because it produces the self-magnetic field and changes the original magnetic field.^{6–8} The impact of a magnetic field and electric fields has numerous applications in engineering, medical field, bio-medical and bio-engineering such as pre-coated metal, magnetic steering of liquid metal mobiles, receiving and transmitting antennas, nuclear reactors cooling, extracting heat stored in the Earth, nonlinear optics, plasma studies, MRI (magnetic resonance imaging), RFA (radiofrequency ablation), drug delivery, blood pumping, cancer therapy, magnetic electro-catalysis, and biosensors.⁹ Kalpana et al.¹⁰ analyzed the Impact of temperature-dependant viscosity and thermal conductivity on MHD boundary layer flow of two-phase dusty fluid through permeable medium. Nadeem et al.¹¹ presented the effect of the magnetic field and partial slip over a stretching sheet with Casson fluid flow. Abbas et al.¹² studied the magnetohydrodynamic flow of dusty fluid over a porous stretching sheet. Acharya et al.¹² analyzed the effect of an inclined magnetic field on hybrid nanofluid over a permeable stretched surface. Ramesh et al.¹³ presented the stagnation point flow on a MHD dusty fluid over a stretching sheet with effect of radiation.

During the last few decades, the study of nanofluids has motivated by numerous researchers, engineers, and scientists due to increases the heat transfer. Nanofluids are produced by dispersing metallic or non-metallic nanometer-size particles in different fluids such as water, ethylene glycol, Bio-fluids, lubricants and engine oils. The term nanofluid was first introduced in 1995 by Choi.¹⁴ Nanofluids have a comprehensive wide range of applications in biomedical, chemical engineering, cancer therapeutics, phase-change material, solar vapor generation, defence sector, heat exchangers, transportation and drug delivery, etc.^{15–18} Khan and Pop¹⁹ investigated numerically in the influence of nanofluids on the boundary layer flow over a stretching flat surface. Pal et al.²⁰ studied the effects of various nanoparticles of Al_2O_3 , Cu and TiO_2 with water on the radiative heat and mass transfer. Devi and Devi²¹ investigated the two-dimensional MHD flow of $Cu - Al_2O_3$ /Water (hybrid nanofluid) over a permeable stretching sheet. Their results observed that the Cu -water nanofluids higher than the under magnetic heat transfer rate of Al_2O_3 /water nanofluids.

The essential objective of the current communication on research is to illustrate the electromagnetohydrodynamic SWCNT-water dusty nanofluid flow near the stagnation-point over a stretching sheet. Both electrical and magnetic field effects are considered for the first time in hybrid dusty fluid flow on the

stagnation point flow. SWCNT-water dusty nanofluid are widely used in wastewater treatment, lunar surface erosion by the exhaust of a landing vehicle, domestic refrigerators, direct absorption solar collectors and many electronic devices in engineering.²² The imperative graphs and tables for the entrenched parameters are drawn to gratify our requirements.

Formulation

Let us consider the steady two-dimensional, laminar boundary layer flow of electromagnetohydrodynamic SWCNT-water dusty nanofluid past over a linear stretching sheet. The allocated sheet is affected by a vertical uniform magnetic field B_0 (see Figure 1(a)). Impact of Joule heating and viscous dissipation are also considered. By keeping the origin fixed, the velocity of the sheet is extended with $u_w = U_0x$, where $U_0 > 0$ is the stretching rate. The temperature of the ambient fluid T_∞ is constant. The surface temperature of the sheet is maintained by convection at a given T_f value. The governing boundary-layer equations for SWCNT-water nanofluid over a sheet can be written as follows.^{3,17}

Fluid phase

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = u_e \frac{du_e}{dx} + v_{nf} \frac{\partial^2 u}{\partial y^2} + \frac{KN}{\rho_{nf}} (u_p - u) + \frac{\sigma_{nf} B_0}{\rho_{nf}} (B_0(u_e - u) + E_0) \quad (2)$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha_{nf} \frac{\partial^2 T}{\partial y^2} + \frac{16\sigma^* T_\infty^3}{3(\rho c_p)_{nf} k^*} \frac{\partial^2 T}{\partial y^2} + \frac{m N c_{pf}}{\tau_t (\rho c_p)_{nf}} (T_p - T) + \frac{m N}{\tau_v (\rho c_p)_{nf}} (u_p - u)^2 + \frac{\mu_{nf}}{(\rho c_p)_{nf}} \left(\frac{\partial u}{\partial y} \right)^2 + \frac{\sigma_{nf}}{(\rho c_p)_{nf}} \{ B_0(u - u_e) - E_0 \}^2 \quad (3)$$

Dust phase

$$\frac{\partial u_p}{\partial x} + \frac{\partial v_p}{\partial y} = 0 \quad (4)$$

$$u_p \frac{\partial u_p}{\partial x} + v_p \frac{\partial u_p}{\partial y} = \frac{K}{m} (u - u_p) \quad (5)$$

$$u_p \frac{\partial T_p}{\partial x} + v_p \frac{\partial T_p}{\partial y} = \frac{c_{pf}}{\tau_t c_{mf}} (T - T_p) \quad (6)$$

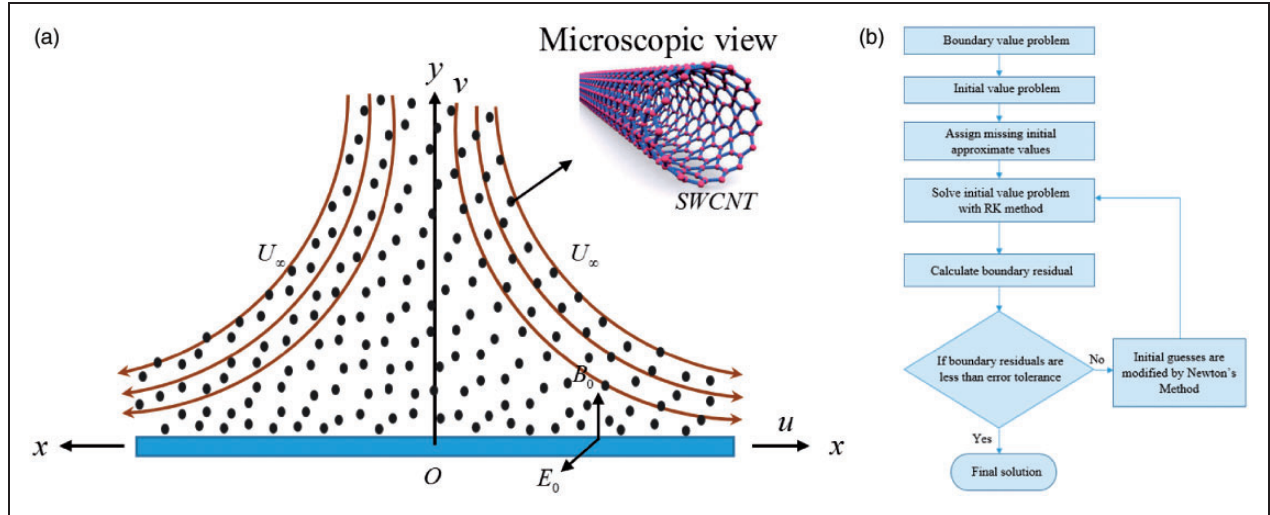


Figure 1. (a) Geometry of the problem; (b) schematic diagram of shooting scheme.

The boundary conditions are³

$$\left. \begin{aligned} u = u_w, \quad v = 0, \quad -k_{nf} \frac{\partial T}{\partial y} = h_f(T_f - T), \quad \text{at } y = 0, \\ u \rightarrow u_e, \quad u_p \rightarrow u_e, \quad v_p \rightarrow v, \quad T \rightarrow T_\infty, \quad T_p \rightarrow T_\infty, \quad \text{as } y \rightarrow \infty. \end{aligned} \right\} \quad (7)$$

The nanofluid quantities are defined as

$$\begin{aligned} \mu_{nf} &= \frac{\mu_f}{(1-\phi)^{2.5}}, \quad v_{nf} = \frac{\mu_{nf}}{\rho_{nf}}, \quad \frac{\rho_{nf}}{\rho_f} = (1-\phi) + \phi \frac{\rho_{CNT}}{\rho_f}, \quad \frac{(\rho c_p)_{nf}}{(\rho c_p)_f} = 1 - \phi + \phi \frac{(\rho c_p)_{CNT}}{(\rho c_p)_f}, \\ \alpha_{nf} &= \frac{k_{nf}}{(\rho c_p)_{nf}}, \quad \frac{\sigma_{nf}}{\sigma_f} = 1 + \frac{3\phi \left(\frac{\sigma_{CNT}}{\sigma_f} - 1 \right)}{\left(\frac{\sigma_{CNT}}{\sigma_f} + 2 \right) - \phi \left(\frac{\sigma_{CNT}}{\sigma_f} - 1 \right)} \\ \frac{k_{nf}}{k_f} &= \frac{(1-\phi) + 2\phi \frac{k_{CNT}}{k_{CNT} - k_f} \ln \frac{k_{CNT} + k_f}{2k_f}}{(1-\phi) + 2\phi \frac{k_f}{k_{CNT} - k_f} \ln \frac{k_{CNT} + k_f}{2k_f}} \end{aligned} \quad (8)$$

Now introducing similarity transformations

$$\begin{aligned} \eta &= \left(\frac{U_0}{v_f} \right)^{0.5} y, \quad \psi = (v_f U_0)^{0.5} x f(\eta), \quad \theta(\eta) = \frac{T - T_\infty}{T_f - T_\infty}, \quad \theta_p(\eta) = \frac{T_p - T_\infty}{T_f - T_\infty}, \quad u = U_0 x f'(\eta), \quad u_p = U_0 x F'(\eta) \\ v &= -(v_f U_0)^{0.5} f(\eta), \quad v_p = -(v_f U_0)^{0.5} F(\eta) \end{aligned} \quad (9)$$

From equations (2) to (9), we get:

Fluid phase

$$\frac{1}{(1-\phi)^{2.5}} f''' + \left((1-\phi) + \phi \frac{\rho_{CNT}}{\rho_f} \right) (f f'' - ((f')^2 - A^2)) + l \beta_v (F' - f') - \frac{\sigma_{nf}}{\sigma_f} M(f' - A - E) = 0 \quad (10)$$

$$\frac{1}{Pr} \left(\frac{k_{nf}}{k_f} + \frac{4}{3} R \right) \theta'' + \frac{(\rho C_p)_{nf}}{(\rho C_p)_f} f \theta' + l \beta_t (\theta_p - \theta) + l Ec \beta_v (F' - f')^2 + \frac{Ec}{(1 - \phi)^{2.5}} f'^2 + M Ec [(f' - A) - E]^2 = 0 \quad (11)$$

Dust phase

$$F F'' - (F')^2 + \beta_v (f' - F') = 0 \quad (12)$$

$$F \theta_p' + \gamma_1 \beta_t (\theta - \theta_p) = 0 \quad (13)$$

With boundary conditions

$$\left. \begin{aligned} f'(0) = 1, f(0) = 0, \quad \theta(0) = \frac{k_{nf}}{k_f Bi} \theta'(0) + 1, \\ f'(\infty) = A, F'(\infty) = A, F(\infty) = f(\infty), \quad \theta(\infty) = 0, \quad \theta_p(\infty) = 0. \end{aligned} \right\} \quad (14)$$

The dimensionless variables are defined as $M = \frac{\sigma B_0^2}{\rho_f U_0}$, $A = \frac{U_\infty}{U_0}$, $Pr = \frac{\mu_f c_p}{k_f}$, $Ec = \frac{u_w^2}{(c_p)_f (T_w - T_\infty)}$, $R = \frac{4\sigma^* T_\infty^3}{3k^* k_f}$, $E = \frac{E_0}{B_0 U_0}$, $\beta_v = \frac{1}{\tau_v U_0}$, $\beta_t = \frac{1}{\tau_t U_0}$, $Bi = \frac{h_f}{k_f} \sqrt{\frac{\nu_f}{U_0}}$ and $l = \frac{mN}{\rho_f}$

Skin friction coefficient, rate of heat transfer are

$$\left. \begin{aligned} C_f Re^{1/2} / 2 &= \frac{1}{(1 - \phi)^{2.5}} f''(0) \\ Nu_x Re^{-1/2} &= - \left(\frac{k_{nf}}{k_f} + \frac{4}{3} R \right) \theta'(0) \end{aligned} \right\} \quad (15)$$

Numerical method and validation

The set of dusty phase and fluid phase differential systems (10)–(13) with the corresponding boundary conditions (14) are solved using Runge Kutta 4th order with shooting method. Let us consider:

$$r_1 = f$$

$$r_1' = r_2$$

$$r_2' = r_3$$

$$r_3' = - \left[(1 - \phi)^{2.5} \left(\left((1 - \phi) + \phi \frac{\rho_{CNT}}{\rho_f} \right) (r_1 r_3 - ((r_2)^2 - A^2)) + l \beta_v (r_7 - r_2) - \frac{\sigma_{nf}}{\sigma_f} M (r_2 - A - E) \right) \right]$$

$$r_4 = \theta$$

$$r_4' = r_5$$

$$r_5' = - \left(\frac{Pr}{\left(\frac{k_{nf}}{k_f} + \frac{4}{3} R \right)} \right) \left(\frac{(\rho C_p)_{nf}}{(\rho C_p)_f} r_1 r_5 + l \beta_t (r_8 - r_4) + l Ec \beta_v (r_7 - r_2)^2 + \frac{Ec}{(1 - \phi)^{2.5}} r_3^2 + M Ec [(r_2 - A) - E]^2 \right)$$

$$r_6 = F$$

$$r_6' = r_7$$

$$r_7' = -\frac{1}{r_6} \left(-(r_7)^2 + \beta_v(r_2 - r_7) \right)$$

$$r_8 = \theta_p$$

$$r_8' = r_9$$

$$r_9' = -\frac{1}{r_6} \left(\gamma_1 \beta_t(r_4 - r_8) \right)$$

with boundary conditions

$$\left. \begin{aligned} r_2(0) = 1, r_1(0) = 0, \quad r_4(0) &= \frac{k_{nf}}{k_f Bi} r_5(0) + 1 \\ r_2(\infty) = A, \quad r_7(\infty) = A, \quad r_6(\infty) &= r_1(\infty), \quad r_8(\infty) = 0, \quad r_9(\infty) = 0 \end{aligned} \right\}$$

Equations (10) to (13) are converted into 1st order differential equations with the help of above expressions, we get

$$\begin{bmatrix} r_1' \\ r_2' \\ r_3' \\ r_4' \\ r_5' \\ r_6' \\ r_7' \\ r_8' \\ r_9' \end{bmatrix} = \begin{bmatrix} r_2 \\ r_3 \\ r_5 \\ -\left[\frac{Pr}{\left(\frac{k_{nf}}{k_f} + \frac{4}{3} R \right)} \right] \left(\frac{(\rho C_p)_{nf}}{(\rho C_p)_f} r_1 r_5 + l \beta_t (r_8 - r_4) + l Ec \beta_v (r_7 - r_2)^2 + \frac{Ec}{(1-\phi)^{2.5}} r_3^2 \right) \\ r_7 \\ -\frac{1}{r_6} \left(-(r_7)^2 + \beta_v(r_2 - r_7) \right) \\ r_9 \\ -\frac{1}{r_6} \left(\gamma_1 \beta_t(r_4 - r_8) \right) \end{bmatrix}$$

With boundary conditions

$$\begin{bmatrix} r_1(0) \\ r_2(0) \\ r_2(\infty) \\ r_4(0) \\ r_6(\infty) \\ r_7(\infty) \\ r_8(\infty) \\ r_9(\infty) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ A \\ \frac{k_{nf}}{k_f Bi} r_5(0) + 1 \\ r_1(\infty) \\ A \\ 0 \\ 0 \end{bmatrix}$$

Runge Kutta 4th order with shooting method algorithm is shown in Figure 1(b).

1×10^{-8} is the error tolerance and $h=0.001$ is the step size of the Runge Kutta 4th order with shooting method algorithm. Thermo-physical properties of the base fluid and nanoparticle is defined in Table 1. The gained consequences of $-\theta'(0)$ values are compared with the earlier published results of Pal²² and Ishak et al.²³ for

Table 1. Electrical conductivity(σ), density(ρ), thermal conductivity(k), specific heat(C_p), and of base fluids and CNT.¹⁵

Physical properties	$C_p \left(\frac{J}{kg K} \right)$	$\rho \left(\frac{kg}{m^3} \right)$	$\sigma \left(\frac{S}{m} \right)$	$k \left(\frac{W}{m K} \right)$
water	4179	997.1	0.05	0.613
SWCNT	796	1600	4.8×10^{-7}	3000

Table 2. Comparison of $-\theta'(0)$ for various values of Pr with previous works of Ishak et al.²³ and Pal.²²

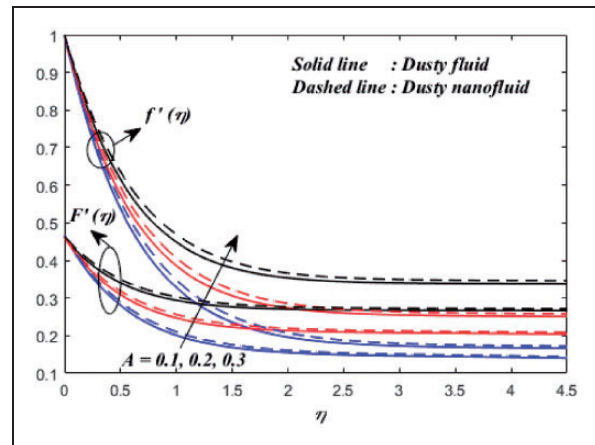
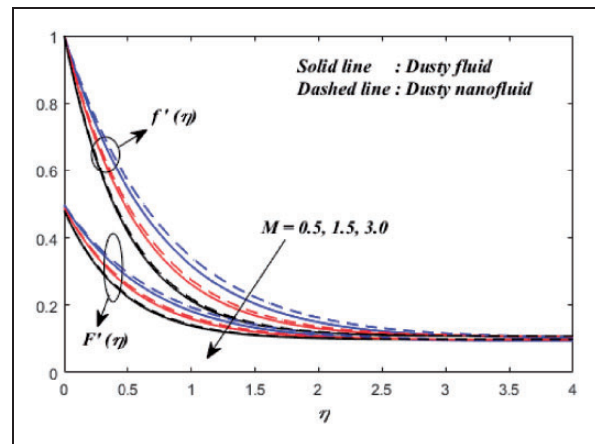
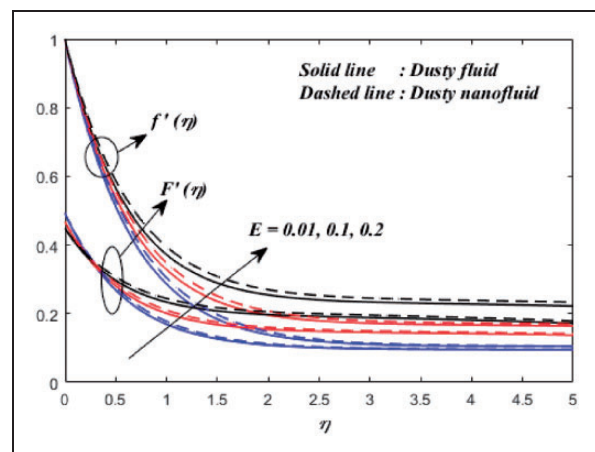
Pr	Ishak et al. ²³ (exact sol.)	Pal ²²	Present results
0.72	0.808631350	0.80863135	0.808631
1.0	1.000000000	1.00000000	1.000000
3.0	1.923682594	1.92368256	1.923683
7.0	3.072250207	3.07225020	3.072250
10.0	3.720673901	3.72067391	3.720674
100.0	12.29408326	12.2940835	12.294083

different values of Pr is exhibited in Table 2. It is shown that the present numerical method is a good agreement with the previous studies.

Results and discussion

This section displays the numerical results for the contour of local Nusselt number and local Sherwood number, as well as the profiles of fluid phase velocity, dusty phase velocity, fluid-phase temperature, dusty phase temperature for dusty fluid and dusty nanofluid case. These results are maintained for the control parameters, i.e. the stagnation parameter (A), fluid particle interaction parameter (β_v), magnetic parameter (M), electric field parameter (E), fluid particle interaction parameter for temperature (β_t), mass concentration of particles (l), Eckert number (Ec), volume fraction of CNT (ϕ) and ratio of specific heat (γ_1).

The computed values of the dusty phase velocity and fluid phase velocity for numerous values of the stagnation parameter (A), magnetic parameter (M) and electric parameter (E) are portrayed in Figures 2 to 4 respectively. Figure 2 establishes the impact of stagnation parameter (A) for dusty fluid and dusty nanofluid cases respectively on the velocity profile. It can be found from these figure that the fluid velocity and dusty fluid velocity increases for larger values of the stagnation parameter (A) for dusty fluid and dusty nanofluid cases. Figure 3 demonstrates the effect of magnetic parameter (M) for dusty fluid and dusty nanofluid cases respectively on the velocity. It is shown that the fluid velocity and dusty fluid velocity decays for higher values of the magnetic parameter (M) for dusty fluid and dusty nanofluid cases respectively. From a physical point of view, as the magnetic field expands, it slows down the fluid by creating resistance in the flow path (Lorentz force). The influence of the electric field parameter (E) on the velocity

**Figure 2.** Effect of A on $f'(\eta)$ and $F'(\eta)$.**Figure 3.** Effect of M on $f'(\eta)$ and $F'(\eta)$.**Figure 4.** Effect of E on $f'(\eta)$ and $F'(\eta)$.

profile for dusty fluid and dusty nanofluid cases shown in Figure 4. It is revealed from this figure the increasing behavior of fluid phase velocity and dusty phase velocity due to increasing the electric field parameter (E) for nanofluid and dusty nanofluid

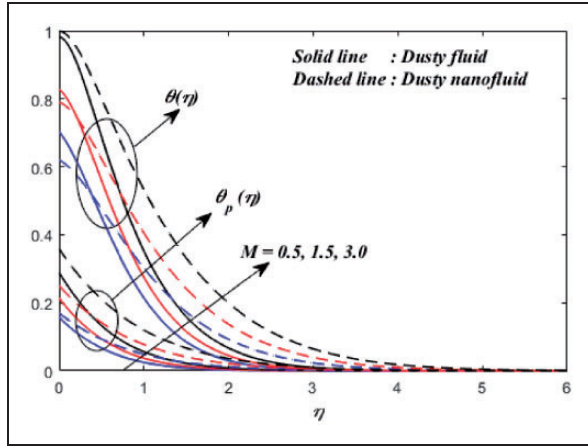


Figure 5. Effect of M on $\theta_p(\eta)$ and $\theta(\eta)$.

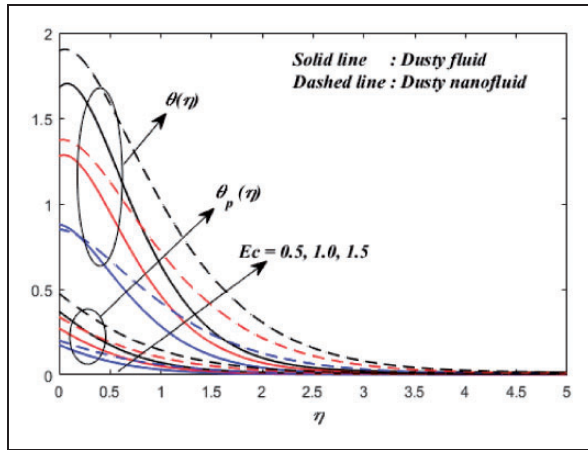


Figure 6. Effect of Ec on $\theta_p(\eta)$ and $\theta(\eta)$.

cases. Physically high values of the electric field parameter give some thermal energy in the flow, which leads to an improvement in the fluid phase velocity and dusty phase velocity.

Figures 5 to 9 show the fluid phase temperature and dusty phase temperature for the various values of the magnetic parameter (M), Eckert number (Ec), stagnation parameter (A), electric field parameter (E), and radiation parameter (R) respectively. Figure 5 presents the impact of magnetic field on fluid phase temperature and dusty phase temperature respectively. It is noticed the fluid phase temperature and dusty phase temperature enhance with expanding magnetic field due to the Lorentz force. Influence of Eckert number (Ec) on the fluid phase temperature and dusty phase temperature for the dusty fluid case and the dusty nanofluid case is plotted in Figure 6. In practically, the dusty phase temperature and fluid phase temperature enhances due to the small effect of the enthalpy difference because the Eckert number $\left(Ec = \frac{u_w^2}{(c_p)_f(T_w - T_\infty)}\right)$ is the kinetic energy flow divided by the difference of an enthalpy within

the boundary layer. This creates the conversion of kinetic energy into internal energy by dissimilar fluid stress. Supplementary Figure 7 shows the variation on the fluid phase temperature and dusty phase temperature with respect to electric field (E). The fluid phase temperature and dusty phase temperature declines with increasing the electric field parameter (E) for nanofluid and dusty nanofluid cases. Actually large values of the electric field parameter lead to an improvement in the fluid phase velocity and dusty phase velocity, which implies fluid phase temperature and dusty phase temperature reduces. From Supplementary Figure 8, one can notice that the fluid phase temperature and dusty phase temperature reduces by an increase of stagnation parameter (A). An increase in the thermal radiation parameter releases thermal energy into the flow, so this energy helps to increase the fluid phase temperature and dusty phase temperature as shown in Supplementary Figure 9. Therefore, higher values of the radiation parameter improve the thermal layer thickness.

The impact of the various parameters the stagnation parameter (A), fluid particle interaction parameter (β_v), electric field parameter (E), fluid particle interaction parameter for temperature (β_t), mass concentration of particles (I) and radiation parameter (R) on skin friction and heat transfer rate for dusty fluid and dusty nanofluid cases has been plotted in Supplementary Figures 10 to 17. Supplementary Figures 10 to 13 shows the impact of the skin friction for the fluid interaction parameter (β_v), electric field parameter (E) and mass concentration of particles (I). It is noticed from this figure the skin friction declines with growing the fluid particle interaction parameter and mass concentration of particles for dusty fluid and dusty nanofluid case, here as reverse trend is observed for electric field parameter (E). Supplementary Figures 14 to 17 illustrate the behavior of radiation parameter (R), the stagnation parameter (A), fluid particle interaction parameter for temperature (β_t) and electric field parameter (E) on the rate of heat transfer. It is observed from this figures the rate of heat transfer increases with the higher values of the radiation parameter (R), fluid particle interaction parameter for temperature (β_t) and the stagnation parameter (A) for both dusty fluid and dusty nanofluid case. Finally, rate of heat transfer is more for $0 \leq E \leq 0.65$ and less for $E > 0.65$.

Conclusion

This article carried out the numerical study on electromagnetic hydrodynamics SWCNT-water dusty nanofluid flow near the stagnation-point over a stretching sheet with the effects of ohmic dissipation, thermal radiation and Joule heating. Major application of SWCNTs in the food industry is packaging. In the food safety area, SWCNTs have potential applications for detecting pathogenic microorganisms by

stabilising antibodies. Furthermore, SWCNTs improves the antimicrobial efficiency of food packaging dependent on cellulose. Some important observations from this analysis are as follows:

- The higher values of the stagnation parameter increase the fluid velocity and dusty fluid velocity.
- When increasing the values of the electric field which enhancing the velocity of the fluid.
- The fluid phase temperature and dusty phase temperature declines with increasing the electric field parameter.
- The skin friction declines with growing the fluid particle interaction parameter and mass concentration of particles for dusty fluid and dusty nanofluid.
- The rate of heat transfer is reduced for the higher values of the fluid particle interaction parameter for temperature.

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Supplemental material

Supplementary material for this article is available online.

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Appendix I

Notation

A	stagnation parameter
B_0	uniform magnetic field (T)
Bi	Biot number
c_{mf}	specific heat of dust particles ($J \cdot K^{-1}$)
c_{pf}	specific heat of fluid particles ($J \cdot K^{-1}$)
E	electric field parameter
E_0	electric field strength ($N \cdot C^{-1}$)
Ec	Eckert number
f	dimensionless fluid velocity
F	dimensionless dusty fluid velocity
h_f	the convective heat transfer ($W \cdot m^{-2} \cdot K^{-1}$)
k	thermal conductivity ($W/m \cdot K$)
k^*	mean absorption coefficient ($W \cdot kg^{-1} \cdot K^{-1}$)
K	Stokes drag constant
l	mass concentration of particles ($kg \cdot m^{-3}$)

m	mass concentration of dust particles ($kg \cdot m^{-3}$)
M	magnetic parameter
N	number density of dust particles
Nu_x	local Nusselt number
Pr	Prandtl number
q_r	radiative heat flux ($W \cdot kg^{-1} \cdot m^{-1}$)
R	radiation parameter
Re_x	local Reynolds number
T	temperature of the fluid (K)
T_f	surface temperature of the sheet (K)
T_p	temperature of the dusty fluid (K)
T_∞	temperature of the ambient fluid (K)
u_e	free stream velocity ($m \cdot s^{-1}$)
u, u_p, v, v_p	Velocity components in x and y directions ($m \cdot s^{-1}$)
μ_{nf}	viscosity of nanofluid ($Ns \cdot m^{-2}$)
α	thermal diffusivity ($m^2 \cdot s^{-1}$)
β_t	fluid particle interaction for temperature
β_v	fluid particle interaction
η	similarity variable
γ_1	ratio of the specific heat
ϕ	nanoparticle volume fraction
τ_t	thermal equilibrium time
τ_v	relaxation time of dust particle
θ	dimensionless temperature of the fluid
θ_p	dimensionless temperature of the dusty fluid
ρ	density ($kg \cdot m^{-3}$)
$(\rho c_p)_f$	heat capacity of fluid particles ($J \cdot K^{-1}$)
σ	electrical conductivity ($S \cdot m^{-1}$)
σ^*	Stefan–Boltzmann constant
ν	kinematic viscosity ($m^2 \cdot s^{-1}$)

Superscript

' differentiation with respect to η

Subscript

CNT	carbon nanotubes
f	fluid particles
nf	nanofluid particles